

FARMBONDHU DATABASE ER DOCUMENTATION

This document summarizes the FarmBondhu PostgreSQL database structure from 'backend/src/db/ensureSchema.js'.

FarmBondhu uses one database for a multi-service SaaS ecosystem:

- Identity and access
- Farm management
- Cow weight and AI feedback
- GreenBondhu Marketplace
- Marketplace chat and moderation
- Community
- VetBondhu veterinary consultation
- MediBondhu human healthcare
- Admin, support, notifications, and operational audit

How To Read This Document

- 'PK' means primary key.
- 'FK' means a declared PostgreSQL foreign key using 'REFERENCES'.
- 'Logical FK' means the column stores another table's ID and is used by the app as a relationship, but the schema does not always declare a PostgreSQL FK constraint.
- 'profiles.id' is the central user identity used by most modules.
- Some domain tables use 'user_id', 'buyer_id', 'seller_id', 'patient_user_id', 'doctor_user_id', or 'vet_user_id'; these are all user identity references to 'profiles.id' unless noted otherwise.

1. Identity And Access

```
erDiagram
    PROFILES {
        uuid id PK
        text email
        text name
        text primary_role
        text status
    }

    AUTH_CREDENTIALS {
        uuid user_id PK_FK
        text password_hash
    }

    USER_ROLES {
        uuid user_id Logical_FK
        text role
    }

    ROLE_PERMISSIONS {
        text role PK
        text permission_code PK
    }
```

```

USER_CAPABILITIES {
  uuid user_id PK_Logical_FK
  text capability_code PK
  boolean is_enabled
  uuid granted_by Logical_FK
}

```

```

APPROVAL_REQUESTS {
  uuid id PK
  uuid user_id Logical_FK
  text request_type
  text status
}

```

```

USER_ADDRESSES {
  uuid id PK
  uuid user_id FK
  text full_name
  text phone
  text division
  text district
  text upazila
}

```

```

PROFILES ||--|| AUTH_CREDENTIALS : auth_credentials
PROFILES ||--o{ USER_ROLES : roles
USER_ROLES }o--o{ ROLE_PERMISSIONS : permissions
PROFILES ||--o{ USER_CAPABILITIES : capabilities
PROFILES ||--o{ USER_CAPABILITIES : granted_by
PROFILES ||--o{ APPROVAL_REQUESTS : requests
PROFILES ||--o{ USER_ADDRESSES : addresses

```

Identity and access table summary

Table	Primary key	FK / logical FK	Purpose
'profiles'	'id'	Root user table	Central identity profile
'auth_credentials'	'user_id'	Declared FK to 'profiles.id'	Password hash for app auth
'user_roles'	unique 'user_id, role'	Logical FK to 'profiles.id'	User role assignment
'role_permissions'	'role, permission_code'	Role lookup	Permission seed map
'user_capabilities'	'user_id, capability_code'	Logical FK to 'profiles.id'; 'granted_by' logical FK	Fine-grained access
'approval_requests'	'id'	Logical FK 'user_id'	Access and role request workflow
'user_addresses'	'id'	Declared FK to 'profiles.id'	Delivery/contact addresses

2. Farm Management And Cow AI

```

erDiagram
    PROFILES {
        uuid id PK
    }

```

```

FARMS {
  uuid id PK
  uuid user_id Logical_FK
}

```

```

text name
text location
text type
}

ANIMALS {
  uuid id PK
  uuid user_id Logical_FK
  uuid farm_id Logical_FK
  text type
  text tracking_mode
  text breed
  text health_status
}

SHEDS {
  uuid id PK
  uuid user_id Logical_FK
  uuid farm_id Logical_FK
  text name
  integer capacity
  text status
}

FEED_RECORDS {
  uuid id PK
  uuid user_id Logical_FK
  uuid farm_id Logical_FK
  uuid animal_id Logical_FK
  date date
  text feed_type
}

FEED_INVENTORY {
  uuid id PK
  uuid user_id Logical_FK
  text name
  numeric stock
}

HEALTH_RECORDS {
  uuid id PK
  uuid user_id Logical_FK
  uuid animal_id Logical_FK
  date date
  text type
}

PRODUCTION_RECORDS {
  uuid id PK
  uuid user_id Logical_FK
  date date
  integer eggs
  numeric milk
}

FINANCIAL_RECORDS {
  uuid id PK
  uuid user_id Logical_FK
  date date
  text type
  numeric amount
}

```

```

MORTALITY_RECORDS {
  uuid id PK
  uuid user_id Logical_FK
  uuid farm_id Logical_FK
  date date
  integer count
}

```

```

SALE_RECORDS {
  uuid id PK
  uuid user_id Logical_FK
  date date
  text product
  numeric total
}

```

```

COW_WEIGHT_ESTIMATIONS {
  uuid id PK
  uuid user_id Logical_FK
  uuid farm_id Logical_FK
  uuid animal_id Logical_FK
  numeric estimated_live_weight_kg
}

```

```

COW_DETECTION_FEEDBACK {
  uuid id PK
  uuid user_id Logical_FK
  uuid estimation_id Logical_FK
}

```

```

PROFILES ||--o{ FARMS : owns
PROFILES ||--o{ ANIMALS : owns
FARMS ||--o{ ANIMALS : contains
FARMS ||--o{ SHEDS : has
PROFILES ||--o{ FEED_RECORDS : logs
FARMS ||--o{ FEED_RECORDS : feed_records
ANIMALS ||--o{ FEED_RECORDS : feed_records
PROFILES ||--o{ FEED_INVENTORY : owns
PROFILES ||--o{ HEALTH_RECORDS : logs
ANIMALS ||--o{ HEALTH_RECORDS : health_records
PROFILES ||--o{ PRODUCTION_RECORDS : logs
PROFILES ||--o{ FINANCIAL_RECORDS : logs
PROFILES ||--o{ MORTALITY_RECORDS : logs
FARMS ||--o{ MORTALITY_RECORDS : mortality
PROFILES ||--o{ SALE_RECORDS : logs
PROFILES ||--o{ COW_WEIGHT_ESTIMATIONS : creates
FARMS ||--o{ COW_WEIGHT_ESTIMATIONS : estimates
ANIMALS ||--o{ COW_WEIGHT_ESTIMATIONS : estimates
COW_WEIGHT_ESTIMATIONS ||--o{ COW_DETECTION_FEEDBACK : feedback
...

```

Farm and AI table summary

Table	Primary key	FK / logical FK	Purpose
---	---	---	---
'farms'	'id'	Logical FK 'user_id'	Farm/workspace records
'animals'	'id'	Logical FK 'user_id', 'farm_id'	Individual or batch animal records
'sheds'	'id'	Logical FK 'user_id', 'farm_id'	Housing/capacity records
'feed_records'	'id'	Logical FK 'user_id', 'farm_id', 'animal_id'	Feed logs
'feed_inventory'	'id'	Logical FK 'user_id'	Feed stock tracking
'health_records'	'id'	Logical FK 'user_id', 'animal_id'	Health/treatment logs

'production_records'	'id'	Logical FK 'user_id'	Milk/egg/output records
'financial_records'	'id'	Logical FK 'user_id'	Income/expense records
'mortality_records'	'id'	Logical FK 'user_id', 'farm_id'	Animal loss records
'sale_records'	'id'	Logical FK 'user_id'	Farm sales records
'cow_weight_estimations'	'id'	Logical FK 'user_id', 'farm_id', 'animal_id'	AI/manual cow weight estimates
'cow_detection_feedback'	'id'	Logical FK 'user_id', 'estimation_id'	Cow detection feedback/training data

3. GreenBondhu Marketplace And Chat

```

erDiagram
    PROFILES {
        uuid id PK
    }

    SHOPS {
        uuid user_id PK_Logical_FK
        text shop_name
    }

    PRODUCTS {
        uuid id PK
        uuid seller_id Logical_FK
        text name
        numeric price
        integer stock
        text listing_status
    }

    ORDERS {
        uuid id PK
        uuid buyer_id Logical_FK
        uuid seller_id Logical_FK
        jsonb items
        numeric total
        text status
    }

    CONVERSATIONS {
        uuid id PK
        uuid buyer_id Logical_FK
        uuid seller_id Logical_FK
        uuid product_id Logical_FK
    }

    CHAT_MESSAGES {
        uuid id PK
        uuid conversation_id Logical_FK
        uuid sender_id Logical_FK
        uuid shared_product_id Logical_FK
    }

    CHAT_MESSAGE_TRANSLATIONS {
        uuid message_id PK_FK
        text target_lang PK
        text translated_text
    }

    CHAT_CONTACT_VIOLATIONS {

```

```

uuid id PK
uuid user_id Logical_FK
uuid conversation_id Logical_FK
}

```

```

MARKETPLACE_CONVERSATION_REPORTS {
  uuid id PK
  uuid conversation_id Logical_FK
  uuid reported_by Logical_FK
  text status
}

```

```

MARKETPLACE_PRODUCT_REVIEWS {
  uuid id PK
  uuid order_id Logical_FK
  uuid product_id Logical_FK
  uuid buyer_id Logical_FK
  uuid seller_id Logical_FK
}

```

```

MARKETPLACE_PRODUCT_COMMENTS {
  uuid id PK
  uuid product_id Logical_FK
  uuid user_id Logical_FK
}

```

```

SELLER_LANE_GRANTS {
  uuid user_id PK_Logical_FK
  text lane PK
  text status
}

```

```

SELLER_WITHDRAWALS {
  uuid id PK
  uuid seller_user_id Logical_FK
  numeric request_amount
  text status
}

```

```

MARKETPLACE_BANNERS {
  uuid id PK
  text image_url
  boolean is_active
}

```

```

MARKETING_DESIGN_DRAFTS {
  uuid id PK
  uuid user_id Logical_FK
  text title
  jsonb canvas_json
}

```

```

PROFILES ||--|| SHOPS : shop
PROFILES ||--o{ PRODUCTS : sells
PROFILES ||--o{ ORDERS : buys
PROFILES ||--o{ ORDERS : sells
PROFILES ||--o{ CONVERSATIONS : buyer
PROFILES ||--o{ CONVERSATIONS : seller
PRODUCTS ||--o{ CONVERSATIONS : discussed
CONVERSATIONS ||--o{ CHAT_MESSAGES : messages
PROFILES ||--o{ CHAT_MESSAGES : sends
PRODUCTS ||--o{ CHAT_MESSAGES : shared
CHAT_MESSAGES ||--o{ CHAT_MESSAGE_TRANSLATIONS : translations

```

```

CONVERSATIONS ||--o{ CHAT_CONTACT_VIOLATIONS : violations
PROFILES ||--o{ CHAT_CONTACT_VIOLATIONS : creates
CONVERSATIONS ||--o{ MARKETPLACE_CONVERSATION_REPORTS : reports
PROFILES ||--o{ MARKETPLACE_CONVERSATION_REPORTS : reports
ORDERS ||--o{ MARKETPLACE_PRODUCT_REVIEWS : reviews
PRODUCTS ||--o{ MARKETPLACE_PRODUCT_REVIEWS : reviews
PRODUCTS ||--o{ MARKETPLACE_PRODUCT_COMMENTS : comments
PROFILES ||--o{ MARKETPLACE_PRODUCT_COMMENTS : writes
PROFILES ||--o{ SELLER_LANE_GRANTS : lane_grants
PROFILES ||--o{ SELLER_WITHDRAWALS : withdrawals
PROFILES ||--o{ MARKETING_DESIGN_DRAFTS : drafts

```

Marketplace table summary

```

| Table | Primary key | FK / logical FK | Purpose |
| --- | --- | --- | --- |
| 'shops' | 'user_id' | Logical FK to 'profiles.id' | Seller shop profile |
| 'products' | 'id' | Logical FK 'seller_id' | Product listings |
| 'orders' | 'id' | Logical FK 'buyer_id', 'seller_id' | Buyer/seller orders |
| 'conversations' | 'id' | Logical FK 'buyer_id', 'seller_id', 'product_id' | Buyer-seller chat threads |
| 'chat_messages' | 'id' | Logical FK 'conversation_id', 'sender_id', 'shared_product_id' | Marketplace chat messages |
| 'chat_message_translations' | 'message_id, target_lang' | Declared FK to 'chat_messages.id' | Per-message translation cache |
| 'chat_contact_violations' | 'id' | Logical FK 'user_id', 'conversation_id' | Contact-guard violations |
| 'marketplace_conversation_reports' | 'id' | Logical FK 'conversation_id', 'reported_by' | Chat report workflow |
| 'marketplace_product_reviews' | 'id' | Logical FK 'order_id', 'product_id', 'buyer_id', 'seller_id' | Product reviews |
|
| 'marketplace_product_comments' | 'id' | Logical FK 'product_id', 'user_id' | Product comments |
| 'seller_lane_grants' | 'user_id, lane' | Logical FK 'user_id', 'reviewed_by' | Seller lane approval |
| 'seller_withdrawals' | 'id' | Logical FK 'seller_user_id', 'reviewed_by' | Seller payout request |
| 'marketplace_banners' | 'id' | Standalone | Browse/banner carousel configuration |
| 'marketing_design_drafts' | 'id' | Logical FK 'user_id' | Seller photo-editor drafts |

```

4. Community And Learning

```

mermaid
erDiagram
    PROFILES {
        uuid id PK
    }

    COMMUNITY_POSTS {
        uuid id PK
        uuid user_id Logical_FK
        text post_type
        text title
        text category
        text status
    }

    COMMUNITY_COMMENTS {
        uuid id PK
        uuid post_id Logical_FK
        uuid user_id Logical_FK
    }

    COMMUNITY_ANSWERS {

```

```

    uuid id PK
    uuid post_id Logical_FK
    uuid user_id Logical_FK
    boolean is_best_answer
}

COMMUNITY_REACTIONS {
    uuid id PK
    uuid post_id Logical_FK
    uuid user_id Logical_FK
}

COMMUNITY_SAVES {
    uuid id PK
    uuid post_id Logical_FK
    uuid user_id Logical_FK
}

COMMUNITY_REPORTS {
    uuid id PK
    uuid post_id Logical_FK
    uuid reported_by Logical_FK
    text status
}

LEARNING_GUIDES {
    uuid id PK
    text title
    text category
    boolean is_published
}

PROFILES ||--o{ COMMUNITY_POSTS : writes
COMMUNITY_POSTS ||--o{ COMMUNITY_COMMENTS : comments
COMMUNITY_POSTS ||--o{ COMMUNITY_ANSWERS : answers
COMMUNITY_POSTS ||--o{ COMMUNITY_REACTIONS : reactions
COMMUNITY_POSTS ||--o{ COMMUNITY_SAVES : saves
COMMUNITY_POSTS ||--o{ COMMUNITY_REPORTS : reports
PROFILES ||--o{ COMMUNITY_COMMENTS : writes
PROFILES ||--o{ COMMUNITY_ANSWERS : writes
PROFILES ||--o{ COMMUNITY_REACTIONS : reacts
PROFILES ||--o{ COMMUNITY_SAVES : saves
PROFILES ||--o{ COMMUNITY_REPORTS : reports
""

```

Community and learning table summary

Table	Primary key	FK / logical FK	Purpose
'community_posts'	'id'	Logical FK 'user_id'	Community posts/questions
'community_comments'	'id'	Logical FK 'post_id', 'user_id'	Post comments
'community_answers'	'id'	Logical FK 'post_id', 'user_id'	Q&A answers
'community_reactions'	'id'; unique 'post_id, user_id'	Logical FK 'post_id', 'user_id'	Reactions
'community_saves'	'id'; unique 'post_id, user_id'	Logical FK 'post_id', 'user_id'	Saved posts
'community_reports'	'id'	Logical FK 'post_id', 'reported_by'	Moderation reports
'learning_guides'	'id'	Standalone	Published learning content

5. VetBondhu

```

""mermaid
erDiagram
    PROFILES {
        uuid id PK
    }

    VETS {
        uuid id PK
        uuid user_id Logical_FK
        text name
        boolean verified
    }

    VET_PROFILES {
        uuid id PK
        uuid user_id Logical_FK
        text full_name
        text verification_status
    }

    VET_AVAILABILITY {
        uuid id PK
        uuid user_id Logical_FK
        text day
        text start_time
        text end_time
    }

    CONSULTATION_BOOKINGS {
        uuid id PK
        uuid patient_mock_id Logical_FK
        uuid vet_id Logical_FK
        uuid vet_user_id Logical_FK
        text status
    }

    CONSULTATION_MESSAGES {
        uuid id PK
        uuid booking_id Logical_FK
        uuid sender_id Logical_FK
    }

    PRESCRIPTIONS {
        uuid id PK
        uuid vet_id Logical_FK
        uuid patient_id Logical_FK
        text status
    }

    PRESCRIPTION_ITEMS {
        uuid id PK
        uuid prescription_id Logical_FK
    }

    E_PRESCRIPTIONS {
        uuid id PK
        uuid patient_mock_id Logical_FK
        uuid vet_id Logical_FK
        text status
    }

    VET_WITHDRAWALS {

```

```

    uuid id PK
    uuid vet_user_id Logical_FK
    numeric request_amount
    text status
}

PROFILES ||--o{ VETS : vet_account
PROFILES ||--o{ VET_PROFILES : vet_profile
PROFILES ||--o{ VET_AVAILABILITY : availability
PROFILES ||--o{ CONSULTATION_BOOKINGS : patient
VETS ||--o{ CONSULTATION_BOOKINGS : assigned
CONSULTATION_BOOKINGS ||--o{ CONSULTATION_MESSAGES : messages
PROFILES ||--o{ CONSULTATION_MESSAGES : sends
VETS ||--o{ PRESCRIPTIONS : issues
PROFILES ||--o{ PRESCRIPTIONS : patient
PRESCRIPTIONS ||--o{ PRESCRIPTION_ITEMS : items
VETS ||--o{ E_PRESCRIPTIONS : issues
PROFILES ||--o{ E_PRESCRIPTIONS : patient
PROFILES ||--o{ VET_WITHDRAWALS : withdrawals
'''

```

VetBondhu table summary

Table	Primary key	FK / logical FK	Purpose
'vets'	'id'	Logical FK 'user_id'	Vet listing/identity
'vet_profiles'	'id'	Logical FK 'user_id'; unique user	Vet verification profile
'vet_availability'	'id'	Logical FK 'user_id'	Vet schedule windows
'consultation_bookings'	'id'	Logical FK 'patient_mock_id', 'vet_id', 'vet_user_id'	VetBondhu consultation booking
'consultation_messages'	'id'	Logical FK 'booking_id', 'sender_id'	VetBondhu room/chat messages
'prescriptions'	'id'	Logical FK 'vet_id', 'patient_id'	Vet prescription header
'prescription_items'	'id'	Logical FK 'prescription_id'	Vet prescription medicines/items
'e_prescriptions'	'id'	Logical FK 'patient_mock_id', 'vet_id'	Active e-prescription records
'vet_withdrawals'	'id'	Logical FK 'vet_user_id', 'reviewed_by'	Vet payout withdrawal

6. MediBondhu

```

'''mermaid
erDiagram
    PROFILES {
        uuid id PK
    }

    MEDIBONDHU_SPECIALTIES {
        uuid id PK
        text name
        text slug
    }

    MEDIBONDHU_HOSPITALS {
        uuid id PK
        text name
        text address
    }

    MEDIBONDHU_DOCTORS {
        uuid id PK
    }

```

```

uuid user_id Logical_FK
uuid specialty_id FK
uuid hospital_id FK
text full_name
text approval_status
}

```

```

MEDIBONDHU_DOCTOR_TIME_SLOTS {
  uuid id PK
  uuid doctor_id FK
  date slot_date
  timestampz slot_start
  timestampz slot_end
}

```

```

MEDIBONDHU_APPOINTMENTS {
  uuid id PK
  uuid patient_user_id Logical_FK
  uuid doctor_id FK
  uuid slot_id FK
  uuid specialty_id FK
  text consultation_type
  text status
}

```

```

MEDIBONDHU_APPOINTMENT_MESSAGES {
  uuid id PK
  uuid appointment_id FK
  uuid sender_id Logical_FK
}

```

```

MEDIBONDHU_PRESCRIPTIONS {
  uuid id PK
  uuid appointment_id FK
  uuid doctor_id FK
  uuid patient_user_id Logical_FK
}

```

```

MEDIBONDHU_PRESCRIPTION_ITEMS {
  uuid id PK
  uuid prescription_id FK
  text medication_name
}

```

```

MEDIBONDHU_DOCTOR_WITHDRAWALS {
  uuid id PK
  uuid doctor_user_id Logical_FK
  numeric request_amount
  text status
}

```

```

MEDIBONDHU_SPECIALTIES ||--o{ MEDIBONDHU_DOCTORS : specialty
MEDIBONDHU_HOSPITALS ||--o{ MEDIBONDHU_DOCTORS : hospital
PROFILES ||--o{ MEDIBONDHU_DOCTORS : doctor_user
MEDIBONDHU_DOCTORS ||--o{ MEDIBONDHU_DOCTOR_TIME_SLOTS : slots
PROFILES ||--o{ MEDIBONDHU_APPOINTMENTS : patient
MEDIBONDHU_DOCTORS ||--o{ MEDIBONDHU_APPOINTMENTS : doctor
MEDIBONDHU_DOCTOR_TIME_SLOTS ||--o{ MEDIBONDHU_APPOINTMENTS : slot
MEDIBONDHU_SPECIALTIES ||--o{ MEDIBONDHU_APPOINTMENTS : specialty
MEDIBONDHU_APPOINTMENTS ||--o{ MEDIBONDHU_APPOINTMENT_MESSAGES : messages
PROFILES ||--o{ MEDIBONDHU_APPOINTMENT_MESSAGES : sends
MEDIBONDHU_APPOINTMENTS ||--o{ MEDIBONDHU_PRESCRIPTIONS : prescription
MEDIBONDHU_DOCTORS ||--o{ MEDIBONDHU_PRESCRIPTIONS : issues

```

```

PROFILES ||--o{ MEDIBONDHU_PRESCRIPTIONS : patient
MEDIBONDHU_PRESCRIPTIONS ||--o{ MEDIBONDHU_PRESCRIPTION_ITEMS : items
PROFILES ||--o{ MEDIBONDHU_DOCTOR_WITHDRAWALS : withdrawals

```

MediBondhu table summary

Table	Primary key	FK / logical FK	Purpose
'medibondhu_specialties'	'id'	Standalone	Human doctor specialties
'medibondhu_hospitals'	'id'	Standalone	Hospital/chamber records
'medibondhu_doctors'	'id'	Declared FK 'specialty_id', 'hospital_id'; logical FK 'user_id'	Human doctor profile
'medibondhu_doctor_time_slots'	'id'	Declared FK 'doctor_id'	Doctor availability slots
'medibondhu_appointments'	'id'	Declared FK 'doctor_id', 'slot_id', 'specialty_id'; logical FK 'patient_user_id'	Patient appointment
'medibondhu_appointment_messages'	'id'	Declared FK 'appointment_id'; logical FK 'sender_id'	Appointment chat
'medibondhu_prescriptions'	'id'	Declared FK 'appointment_id', 'doctor_id'; logical FK 'patient_user_id'	Human prescription header
'medibondhu_prescription_items'	'id'	Declared FK 'prescription_id'	Prescription items
'medibondhu_doctor_withdrawals'	'id'	Logical FK 'doctor_user_id', 'reviewed_by'	Doctor payout withdrawal

7. Admin, Support, Notifications, And Audit

```

mermaid
erDiagram
    PROFILES {
        uuid id PK
    }

    NOTIFICATIONS {
        uuid id PK
        uuid user_id Logical_FK
        text title
        text type
        boolean read
    }

    ADMIN_TEAM {
        uuid id PK
        uuid user_id Logical_FK
        text email
        text admin_role
    }

    EMAIL_AUDIT_LOG {
        uuid id PK
        text email_type
        text recipient_email
        text status
    }

    REGISTRATION_PENDING {
        text email PK
        jsonb profile_data
        timestampz expires_at
    }

    PASSWORD_RESET_PENDING {

```

```

text email PK
uuid user_id Logical_FK
timestamp expires_at
}

```

```

PROFILES ||--o{ NOTIFICATIONS : receives
PROFILES ||--o{ ADMIN_TEAM : admin_profile
PROFILES ||--o{ PASSWORD_RESET_PENDING : resets
...

```

Admin/support table summary

Table	Primary key	FK / logical FK	Purpose
'notifications'	'id'	Logical FK 'user_id'	Account-scoped notifications
'admin_team'	'id'	Logical FK 'user_id'	Admin team profiles/permissions
'email_audit_log'	'id'	Standalone	Email delivery/audit records
'registration_pending'	'email'	Standalone	OTP registration staging
'password_reset_pending'	'email'	Logical FK 'user_id'	Password reset OTP staging

Notes About Declared Vs Logical Foreign Keys

The schema declares some FK constraints directly, especially in:

- 'auth_credentials.user_id -> profiles.id'
- 'user_addresses.user_id -> profiles.id'
- 'chat_message_translations.message_id -> chat_messages.id'
- MediBondhu doctor, slot, appointment, message, and prescription tables

Many other relationships are enforced by application logic and ownership checks rather than database-level constraints. This is why the diagrams label them as 'Logical_FK'.

For example:

- 'farms.user_id' logically points to 'profiles.id'
- 'animals.farm_id' logically points to 'farms.id'
- 'orders.buyer_id' and 'orders.seller_id' logically point to 'profiles.id'
- 'community_comments.post_id' logically points to 'community_posts.id'
- 'consultation_messages.booking_id' logically points to 'consultation_bookings.id'

Recommended Future Schema Hardening

If the project moves toward stricter database integrity, consider adding declared FK constraints for high-value relationships:

- Farm records to 'profiles', 'farms', and 'animals'
- Marketplace orders/products/chats to users and products
- Community comments/answers/reactions/saves to posts and users
- VetBondhu bookings/messages/prescriptions to users and vets
- Notification ownership to 'profiles'

Before adding strict constraints, clean or backfill existing data so older rows do not violate the new rules.